programm 4

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4. Develop a script in R to produce a bar graph displaying the frequency distribution of categorical data in a given dataset , grouped by a specific variable, using ggplot2.

step 1:Load necessary library

```
library(ggplot2)
```

step 1: Load the Dataset

```
data <- mtcars
head(data)
```

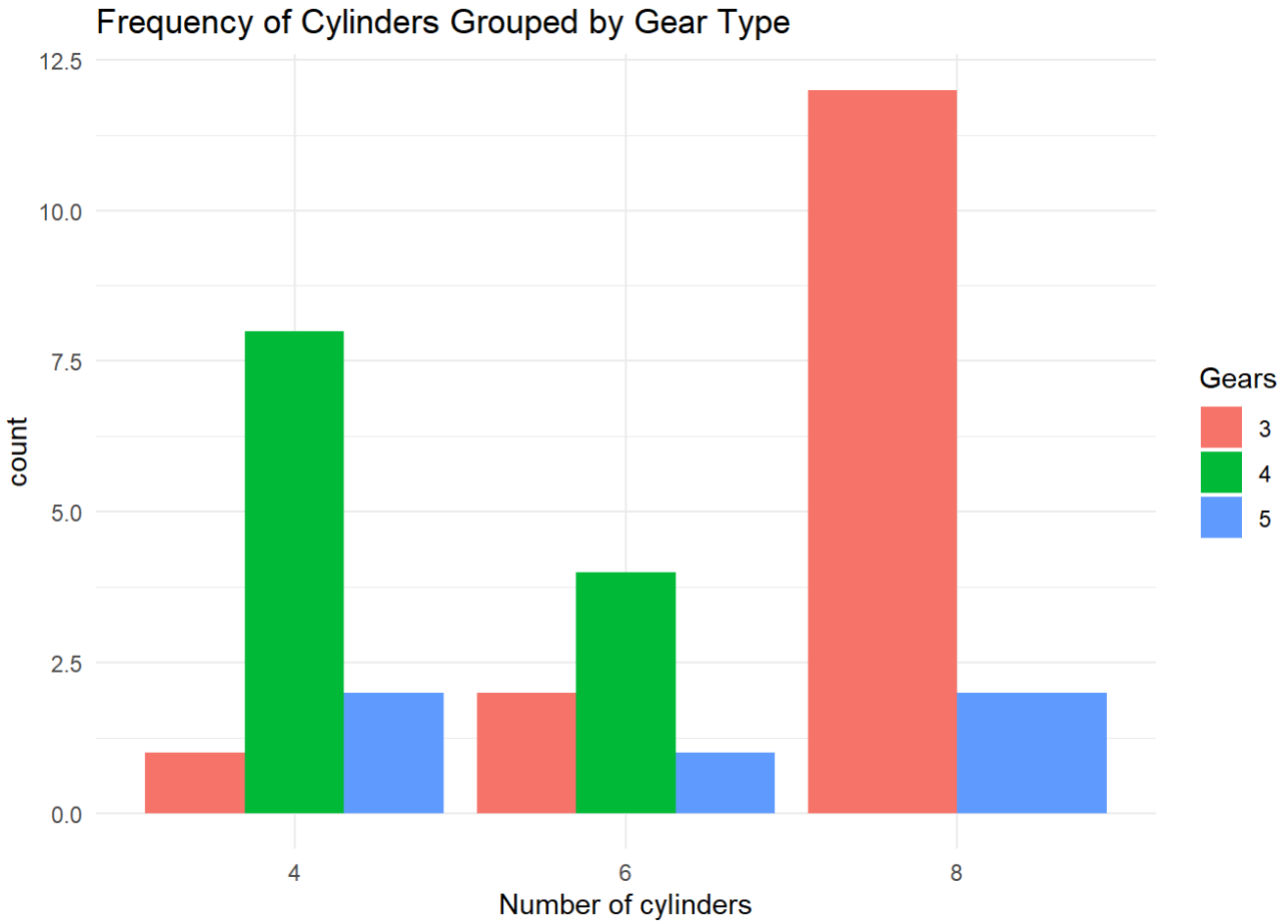
```
                   mpg cyl disp  hp drat    wt  qsec vs am gear carb
Mazda RX4         21.0   6  160 110 3.90 2.620 16.46  0  1    4    4
Mazda RX4 Wag     21.0   6  160 110 3.90 2.875 17.02  0  1    4    4
Datsun 710        22.8   4  108  93 3.85 2.320 18.61  1  1    4    1
Hornet 4 Drive    21.4   6  258 110 3.08 3.215 19.44  1  0    3    1
Hornet Sportabout 18.7   8  360 175 3.15 3.440 17.02  0  0    3    2
Valiant           18.1   6  225 105 2.76 3.460 20.22  1  0    3    1
```

step 2: Convert Numeric Data to Categorical

```
data$cyl <- as.factor(data$cyl)
data$gear <- as.factor(data$gear)
```

step 3: Create a bar graph

```
ggplot(data,aes(x =cyl, fill = gear))+
  geom_bar(position = "dodge") +
  labs(title = "Frequency of Cylinders Grouped by Gear Type",
       x = "Number of cylinders",
       y = "count",
       fill = "Gears")+
  theme_minimal()
```


Frequency of Cylinders Grouped by Gear Type
12.5
10.0
7.5
5.0
2.5
0.0
count
4
6
8
Number of cylinders
Gears
3
4
5